

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I am looking for a HV connector for a wire harness material

Assistant: To better understand your requirements, could you please clarify:

1. What is the exact use of the polymer in your HV connector application? For example, is it for the connector insert, housing, cover, or harness protector?
2. Where will this HV connector be used in your solution? Is it for automotive (e.g., electric vehicle battery systems, under-hood wiring), industrial equipment, or another end-use?

Your answers will help me identify the most suitable material options for your application.

User: this is for an EV battery, housing

Assistant: Thank you for clarifying. To proceed, could you please provide:

1. What are the typical operating conditions for your EV battery connector housing? For example, is it located inside the battery compartment (protected from direct exposure), or is it exposed to under-hood conditions (heat, vibration, moisture, automotive fluids)?
2. What is the minimum wall thickness of your connector housing part?
3. Do you have any preferred base materials for this application (such as PA66, PBT, PPS, etc.), or are you open to alternative material suggestions?
4. If you have a preferred material, are you open to considering alternatives if they offer

better performance or cost-effectiveness?

Your answers will help ensure the material suggestions are tailored to your specific requirements.

User: thickness of 0.8mm and i dont have any preference

Assistant: Here are the key material requirements (CTQs) assumed for your HV connector housing application:

1. Minimum wall thickness: 0.8 mm
2. Flame rating: UL94 V-0 at 0.8 mm
3. Comparative Tracking Index (CTI): 600 V
4. Relative Thermal Index - Electrical (RTI Elec): $\geq 130^{\circ}\text{C}$
5. Glow Wire Flammability Index (GWFI): 960°C at 0.8 mm
6. Glow Wire Ignition Temperature (GWIT): $\geq 850^{\circ}\text{C}$ at 0.8 mm
7. Hot Wire Ignition (HWI): PLC 0-1 at 0.8 mm
8. Dielectric strength: ≥ 20 kV/mm
9. Flame retardant and hydrolysis resistant
10. Injection molding suitability

Are all of these requirements correct for your application? Please confirm so I can proceed to refine the material selection.

User: neec plc 0

Assistant: Thank you for confirming PLC 0 for Hot Wire Ignition (HWI) at 0.8 mm. Next, I need your input on a few additional requirements to ensure the material is fully suitable:

1. Glass fiber content (Filler): Do you have a target or preference for glass fiber percentage (e.g., GF15, GF25, GF30)?
2. Do you require orange color stability for HV identification, or is color not important for your application?
3. Are there any specific fluids (coolants, battery electrolyte, cleaners) that the connector housing needs to resist?

Please let me know your preferences or requirements for these points.

User: gf and color is not important

Assistant: Thank you for clarifying. Could you please confirm or provide details for the following:

1. Are there any specific fluids (such as coolants, battery electrolyte, or cleaners) that the connector housing needs to resist?
2. Is compatibility with sealing materials (such as elastomeric gaskets or O-rings) required for your housing?
3. Are there any OEM-specific tests or standards (like LV124, GMW, etc.) that the material must meet?

Your input will help ensure the material suggestions are fully aligned with your application needs.

User: no need suggest grades on the ctq we discussed so far

Assistant: Here are potential grade options for your HV connector housing based on the requirements discussed:

1. Zytel® HTNFR42G30NH NC010 – Meets most CTQs including UL94 V-0, CTI 600 V, RTI

Electrical 130°C, GWFI 960°C, GWIT 800°C at 0.8 mm, high dielectric strength, hydrolysis resistance, and injection molding suitability. Note: GWIT is slightly below your target (800°C vs. 850°C), and HWI at 0.8 mm is PLC 3.

2. CELANEX® XFR 6852 GF30 – Meets UL94 V-0, CTI 600 V, RTI Electrical 140°C, GWFI 960°C, GWIT 775–800°C at 0.8 mm, high dielectric strength, hydrolysis resistance, and injection molding suitability. Note: GWIT is below target, and HWI PLC 0 is at ≥ 1.5 mm.

3. Zytel® HTNFR52G30BL NC010 – Offers very high GWIT (925°C at 0.75 mm), RTI Electrical 140°C, UL94 V-0, and HWI PLC 0, but CTI is lower (PLC 1, 450–525 V).

4. Zytel® FR95G25V0NH NC010 – Meets UL94 V-0, CTI 600 V, RTI Electrical 160°C, GWFI 960°C, HWI PLC 0 at ≥ 0.75 mm, high dielectric strength, hydrolysis resistance, and injection molding suitability. GWIT is 700–725°C at 0.75–3.0 mm.

5. FORTRON® 6162A7 – Meets GWIT $\geq 850^\circ\text{C}$ and RTI Electrical 220°C, but CTI is much lower (PLC 3, 175–249 V) and dielectric strength is ~ 12 kV/mm.

These grades may be suitable options, but please review the trade-offs, especially regarding GWIT and HWI at 0.8 mm. Final material selection and product design decisions rest with you. Ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.